ZIKEY 字鑰

Which character should I learn next—and why?

A connected map, visible evidence,
and a learning order you can steer.



Functional local prototype

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Information Visualization · NCKU

Data, AI, visualization and learner feedback

DATA

Preserved snapshots

TOCFL + MMH + CC-CEDICT

AI

Score + local explanation

NetworkX + Ollama

VISUALIZATION

Three linked views

Streamlit + D3 + Plotly

HUMAN

Select, steer, mark known

One shared learner state

One traceable transformation

TOCFL: 請問 / qǐngwèn / Novice 1 / source row

→ qǐngwèn · {請, 問} · “Excuse me, may I ask...?” · source retained

Real data, with visible quality limits

290 learning characters · 450 words · 556 IDS leaf edges

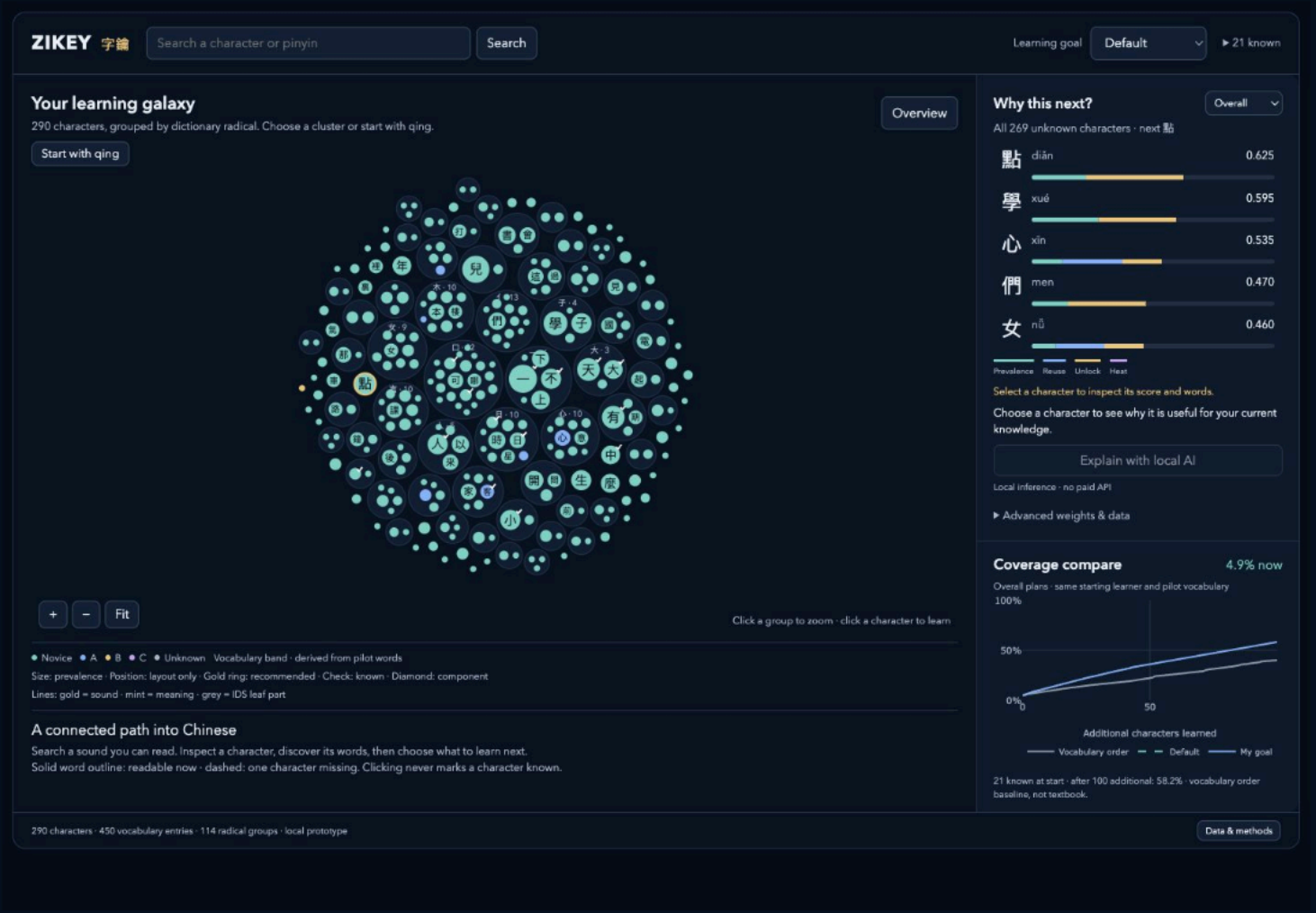
Source	Fields used	Version / terms
TOCFL	Traditional words, reading, level, POS	2024-09-23 workbook
Make Me a Hanzi	Radical, IDS, available origin hints	SHA-256 snapshot / LGPL
CC-CEDICT	English senses and readings	2026-11-04 / CC BY-SA 4.0

6 missing glosses / 84 senses need review / 30 missing origin records

Vocabulary bands are derived from pilot words. Missing data stays labelled.

Sources: SC-TOP (2024), skishore (n.d.), CC-CEDICT contributors (2026)

Explore the overview, then focus on learning



Colour

Derived vocabulary band

Size

Pilot vocabulary prevalence

Ring / check

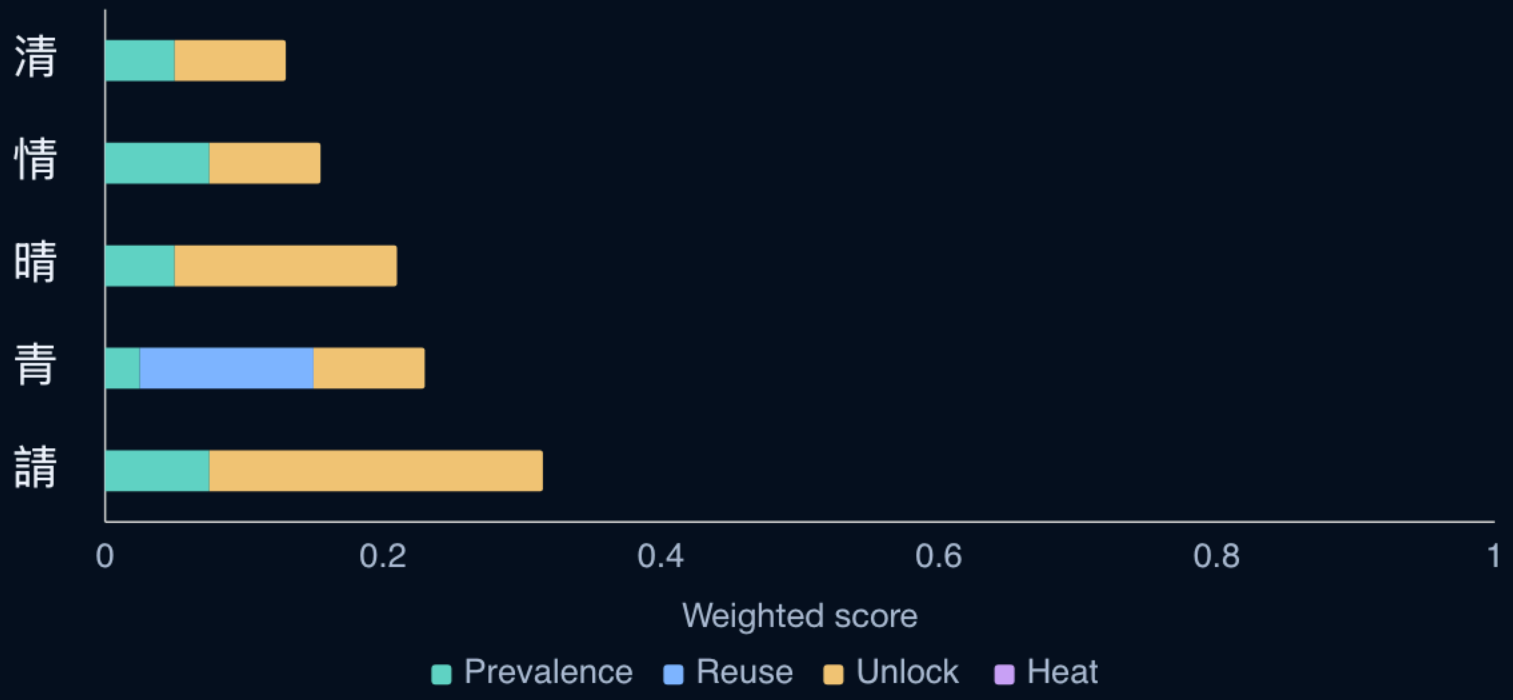
Recommended / known

Position

Packing layout only

Recommendations expose their evidence

Default: prevalence 35% · reuse 25% · unlock 40% · heat 0%



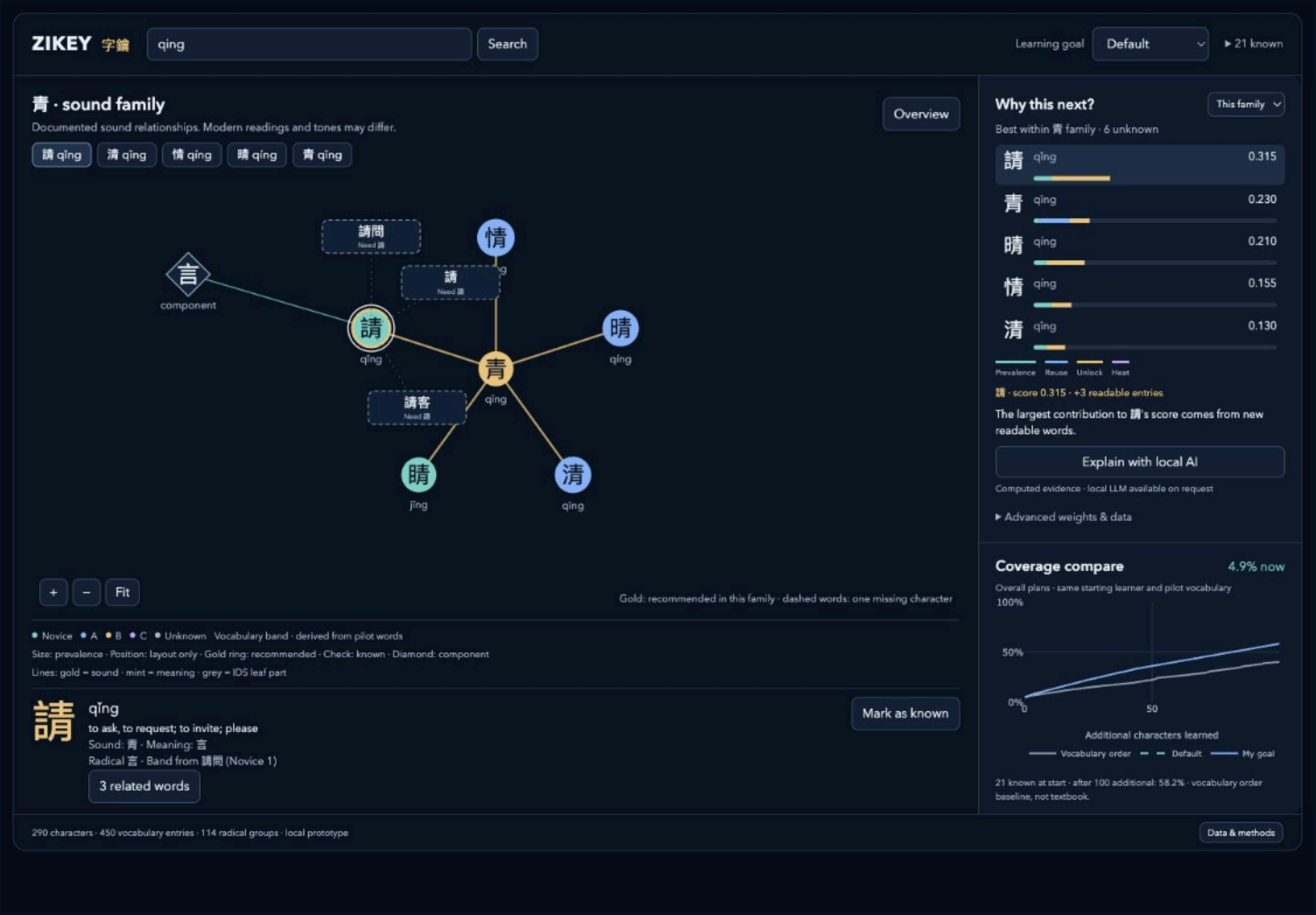
One missing character

請問

Learning 請 unlocks this word only if 問 is known.

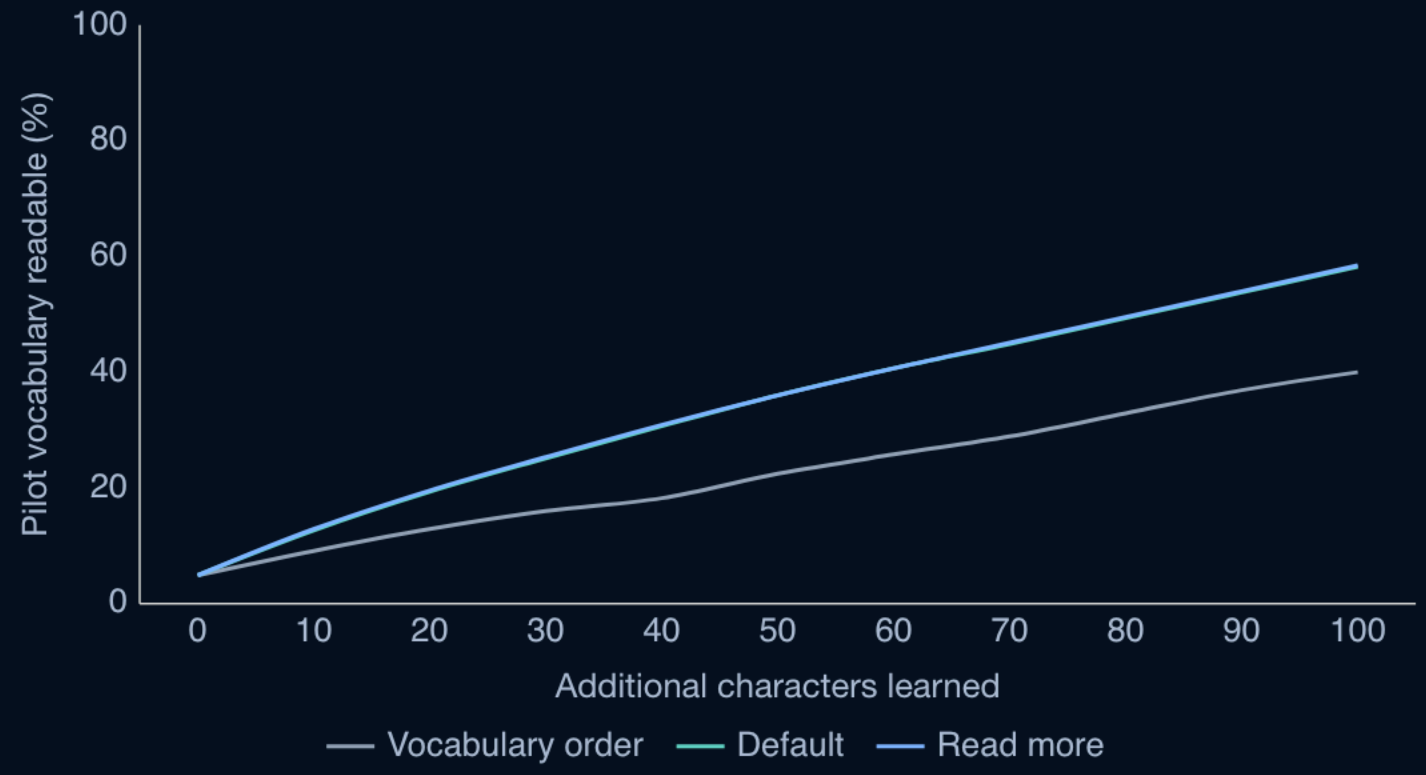
Local AI selects supported explanations. The visible formula determines the ranking.

Live task: choose and learn the next character



1. Search qing → select 請
2. Inspect parts + 請問
3. Explain; change the goal
4. Mark known; see updates

What has been verified



22 automated tests passed

Linked browser task verified
5 supported local-AI checks
Fresh requests: 5.5–13.6 s

21 → 22 known characters
4.9% → 5.6% readable words

In-sample vocabulary simulation. Learning effectiveness remains untested.

Roadblocks, decisions and next work

Roadblock	Implemented decision	Next evidence
Ambiguous / missing language fields	Source-backed matches; visible gaps	Review readings and 84 senses
Crowded graph and linked state	Overview → focus; separate edge types	T1–T3 tasks with 3–5 learners
Local AI latency / unsupported prose	Cache + constrained selection + fallback	Measure more devices and requests
Coverage proxy and curated subset	Same words / start; honest baseline	Corpus frequency; held-out evaluation

Deferred: ALS, accounts, cloud hosting, full 10,000-node graph and learning-effect claims

References and AI disclosure

CC-CEDICT contributors. (2026). *CC-CEDICT* [Data set]. MDBG. Retrieved November 4, 2026, from

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Hanzi Constellation. (n.d.). *Explore* [Interactive visualization]. Retrieved November 4, 2026, from

https://hanzi-constellation.app/explore?c=青

skishore. (n.d.). *Make Me a Hanzi* [Data set]. GitHub. Retrieved November 4, 2026, from

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Steering Committee for the Test Of Proficiency–Huayu. (2024). *華語八千詞表* [Chinese vocabulary list].

https://tocfl.edu.tw/tocfl/assets/files/vocabulary/8000zhuyin_202409.zip

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Development: Codex assisted code, tests and slides. Runtime: local Qwen selects supplied evidence.